EULER CHARACTERISTIC OF TWISTS IS A NUMERICAL POLYNOMIAL OF DEGREE AT MOST THE SUPPORT DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Dependency disclosure: the closing step 'A simple arithmetic argument shows' summarizes the standard finite-difference characterization of numerical polynomials; every preceding reduction is fully spelled in the span.

Allowed standard prerequisites (editorial).

- delivered companions: additivity and chi on finite support. delivered companions lemma-euler-characteristic-additive, lemma-chi-tensor-finite. Additivity in short exact sequences; constant base case.
- Divisors toolbox (statement level, cross-chapter). Divisors, Lemmas divisors-lemma-remove-embedded-points, divisors-lemma-no-embedded-points-endos, divisors-lemma-regular-meromorphic-section-exists, divisors-lemma-make-maps-regular-section. Embedded-point removal, support normalization, regular meromorphic sections and the regular maps.
- Cohomology of Schemes, affine-hypercover cohomology agreement. Cohomology of Schemes, Lemma coherent-lemma-relative-affine-cohomology. chi agreement along closed immersions.

Source reference. Original lemma 0AYT: sources/stacks-chow/tags.txt, source lines 13801–13801. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02OL: sources/stacks-chow/tags.txt, source lines 3520–3520. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08AA: sources/stacks-chow/tags.txt, source lines 10525–10525. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 089W: sources/stacks-chow/tags.txt, source lines 10511–10511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02OM: sources/stacks-chow/tags.txt, source lines 3521–3521. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02OZ: sources/stacks-chow/tags.txt, source lines 3534–3534. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 02P2: sources/stacks-chow/tags.txt, source lines 3537–3537. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Let $\mathcal{L}_1, \ldots, \mathcal{L}_r$ be invertible $\mathcal{O}_X$-modules. The map*

$$(n_1, \ldots, n_r) \longmapsto \chi(X, \mathcal{F} \otimes \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r})$$

is a numerical polynomial in $n_1, \ldots, n_r$ of total degree at most the dimension of the support of $\mathcal{F}$.

Proof. We prove this by induction on $\dim(\mathrm{Supp}(\mathcal{F}))$. If this number is zero, then the function is constant with value $\dim_k \Gamma(X, \mathcal{F})$ by Lemma 0AYT. Assume $\dim(\mathrm{Supp}(\mathcal{F})) > 0$.

If $\mathcal{F}$ has embedded associated points, then we can consider the short exact sequence $0 \to \mathcal{K} \to \mathcal{F} \to \mathcal{F}' \to 0$ constructed in Divisors, Lemma 02OL. Since the dimension of the support of $\mathcal{K}$ is strictly less, the result holds for $\mathcal{K}$ by induction hypothesis and with strictly smaller total degree. By additivity of the Euler characteristic (Lemma 08AA) it suffices to prove the result for $\mathcal{F}'$. Thus we may assume $\mathcal{F}$ does not have embedded associated points.

If $i : Z \to X$ is a closed immersion and $\mathcal{F} = i_*\mathcal{G}$, then we see that the result for X, $\mathcal{F}$, $\mathcal{L}_1, \ldots, \mathcal{L}_r$ is equivalent to the result for Z, $\mathcal{G}$, $i^*\mathcal{L}_1, \ldots, i^*\mathcal{L}_r$ (since the cohomologies agree, see Cohomology of Schemes, Lemma 089W). Applying Divisors, Lemma 02OM we may assume that X has no embedded components and $X = \mathrm{Supp}(\mathcal{F})$.

Pick a regular meromorphic section s of $\mathcal{L}_1$, see Divisors, Lemma 02OZ. Let $\mathcal{I} \subset \mathcal{O}_X$ be the ideal of denominators of s and consider the maps

$$\mathcal{I}\mathcal{F} \to \mathcal{F}, \quad \mathcal{I}\mathcal{F} \to \mathcal{F} \otimes \mathcal{L}_1$$

of Divisors, Lemma 02P2. These are injective and have cokernels $\mathcal{Q}$, $\mathcal{Q}'$ supported on nowhere dense closed subschemes of $X = \mathrm{Supp}(\mathcal{F})$. Tensoring with the invertible module $\mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r}$ is exact, hence using additivity again we see that

$$\begin{aligned} &\chi(X, \mathcal{F} \otimes \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r}) - \chi(X, \mathcal{F} \otimes \mathcal{L}_1^{\otimes n_1 + 1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r}) \\ &= \chi(\mathcal{Q} \otimes \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r}) - \chi(\mathcal{Q}' \otimes \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_r^{\otimes n_r}) \end{aligned}$$

Thus we see that the function $P(n_1, \ldots, n_r)$ of the lemma has the property that

$$P(n_1 + 1, n_2, \ldots, n_r) - P(n_1, \ldots, n_r)$$

is a numerical polynomial of total degree $<$ the dimension of the support of $\mathcal{F}$. Of course by symmetry the same thing is true for

$$P(n_1, \ldots, n_{i-1}, n_i + 1, n_{i+1}, \ldots, n_r) - P(n_1, \ldots, n_r)$$

for any $i \in \{1, \ldots, r\}$. A simple arithmetic argument shows that P is a numerical polynomial of total degree at most $\dim(\mathrm{Supp}(\mathcal{F}))$. □